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**2023-3R/2024 (MAT3104C)**

**2024**

# MATHEMATICS

Paper : MAT3104C

***( Ordinary Differential Equation )***

*Full Marks : 45*

Time : 1½ hours

***The figures in the margin indicate full marks for the questions.***

1. Answer the following : 2×5=10

(a) Solve $xydx + (x+1)dy = 0$

(b) Test for exactness of the differential equation $(1+y)dx + (1-x)dy = 0$

(c) Obtain the integrating factor (I.F) of the differential equation

$$\left(1-x^2\right)\frac{dy}{dx} - xy = 1$$

*Contd.*

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(d) If $y_1(x) = \sin 3x$ and $y_2(x) = \cos 3x$ are two solutions of the differential equation $\frac{d^2y}{dx^2} + 9y = 0$, then show that $y_1(x)$ and $y_2(x)$ are linearly independent.

(e) Find the particular integral(P.I) of the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = e^x \sin x$$

2. Answer the following questions : 5×3=15

(a) Solve $\frac{x\,dy}{dx} + y = x^4y^3$

**Or**

Solve $\left(1-x^2\right)\frac{dy}{dx} + 2xy = x\sqrt{1-x^2}$

(b) Solve by the method of variation of parameters

$$x^2\frac{d^2y}{dx^2} + x.\frac{dy}{dx} - y = x^2e^x$$



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(c) Solve

$$x^2D^2y - 3xDy + 5y = x^2 \sin\log x$$

***Or***

$$4xp^2 = (3x-a)^2$$

3. Answer ***either*** [(a) and (b)] ***or*** [(c) and (d)] : 5+5=10

(a) Solve by the method of undetermined co-efficients

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = xe^x$$

(b) Solve $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} + 4y = \sin 2x$

subject to the conditions $y = \dfrac{1}{8}$ and $Dy = 4$ when $x = 0$.

(c) Test for exactness and hence solve the differential equation

$$\left(y^4 + 4x^3y + 3x\right)dx + \left(x^4 + 4xy^3 + y + 1\right)dy = 0$$



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(d) Find the interpreting factor (I.F) and solve

$$y\left(xy+2x^2y^2\right)dx+x\left(xy-x^2y^2\right)dy=0$$

4. Answer **either** [(a) and (b)] **or** [(c) and (d)] : 5+5=10

(a) Use Wronskian to show that the function $x, x^2, x^3$ are linearly independent. Determine the differential equation with these as the independent solutions.

(b) A population grows at the rate of 5% in a year. How long does it take for the population to double ?

(c) If $y=u$ is a solution of $y''+P(x)y'+Q(x)y=0$, then show that another solution $y=\nu$ of the solution is given by

$$\nu=u\int\frac{W(u,\nu)}{u^2}\,dx$$

where the Wronskian $W(u,\nu)$ satisfies the differential equation

$$\frac{d}{dx}W(u,\nu)+PW(u,\nu)=0$$



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(d) A bacteria is known to grow at a rate proportional to the amount present. After one hour 1000 strands of bacteria are observed in the culture, after 4 hours 3000 strands of bacteria are found. Find

(i) an expression for the number of strands of the bacteria present in the culture of any time $t$.

(ii) the number of strands of the bacteria originally in the culture.

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